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Statistical Models
Week 2: stochastic convergence and properties of mle

We will continue with modes of stochastic convergence, recall the law of large numbers and the central limit theorem as examples, and see how these modes of convergence relate to each other.

Under certain conditions, the maximum likelihood estimator (mle) is consistent and asymptotically normal distributed. In context with the proof of consistency of mle we will shortly come across the Kullback-Leibler-divergence, a way to characterize the distance between two distributions that which will be important later in the course.

On the way to proving the asymptotic normality of MLE, we will take a closer look at the score function $U(\theta; Y)$, and derive Bartlett's identities about the mean and variance of $U(\theta; Y)$.

Relevant Readings

Davison (2003), Sections 2.2.1, 4.4. (4.6 for counterexamples)

Lecture Material

[4]: for lecture 4 (1st lecture this week), [5] for lecture 5, [4+5] both

- [4+5] Link to onenotes on Google drive
- [4] Examples for stochastic convergence
- [4+5] Experiment: asymptotic behaviour of mle

Exercises 2

You need not have solved all problems before the exercise class. However to get most out of the class, read the problems thoroughly and try at least some of them before the theoretical exercises.

Problem 2.1: Convergence of the product of two sample means

Davison(2003), Section 2.2: Exercise 4 (slightly extended)
Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ be independent random samples from the exponential densities $f_X = \lambda e^{-\lambda x},\ x > 0$ and $f_Y(y) = \lambda^{-1} e^{-y/\lambda},\ y > 0$, with $\lambda > 0$. Let $\overline{X}$ and $\overline{Y}$ denote the sample averages.

a) Show that $\overline{XY} \xrightarrow{P} 1$ as $n \to \infty$.

b) Write a simulation to illustrate the result in a). You can do this even if you don't think you succeeded with a formal proof; use a programming language of your choice.

Problem 2.2: Maximum likelihood and information in a linear model

Davison (2003) Section 4.9: Problem 10.

In a normal linear model through the origin, independent observations $Y_1, \ldots, Y_n$ are such that $Y_j \sim N(\beta x_j, \sigma^2)$.
Show that the log likelihood for a sample $y_1, \ldots, y_n$ is

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{j=1}^{n}(y_j - \beta x_j)^2.$$

Deduce that the likelihood equations are equivalent to $\sum x_j(y_j - \hat{\beta}x_j) = 0$ and $\hat{\sigma^2} = n^{-1}\sum(y_j - \hat{\beta}x_j)^2$, and hence find the maximum likelihood estimates $\hat{\beta}$ and $\hat{\sigma^2}$ for data with $x = (1, 2, 3, 4, 5)$ and $y = (2.81, 5.48, 7.11, 8.69, 11.28)$.
Show that the observed information matrix evaluated at the maximum likelihood estimates is diagonal and use it to obtain approximate 95% confidence intervals for the parameters. Plot the data and your fitted line $y = \hat{\beta}x$. Say whether you think the model is correct, with reasons. Discuss the adequacy of the normal approximations in this example.

Three small exercises covering facts and examples for convergence in probability:

Problem 2.3: Continuous mapping theorem, version for convergence in probability

Davison (2003) Section 2.2, Exercise 1.
Suppose that $S_n \xrightarrow{P} s_0$, and that the function h is continuous at s_0, that is, for any $\varepsilon > 0$ there exists a $\delta > 0$ such that $|x - y| < \delta$ implies that $|h(x) - h(y)| < \varepsilon$. Explain why this implies that

$$\mathrm{P}(|S_n - s_0| < \delta) \leq \mathrm{P}(|h(S_n) - h(s_0)| < \varepsilon) \leq 1,$$

and deduce that $\mathrm{P}(|h(S_n) - h(s_0)| < \varepsilon) \longrightarrow 1$ as $n \to \infty$. That is, $h(S_n) \xrightarrow{P} h(s_0)$.

Problem 2.4: Convergence in distribution to a constant

Davison (2003) Section 2.2, Exercise 2.
Let s_0 be a constant. By writing

$$\mathrm{P}(|S_n - s_0| \leq \varepsilon) = \mathrm{P}(S_n \leq s_0 + \varepsilon) - \mathrm{P}(S_n \leq s_0 - \varepsilon),$$

for $\varepsilon > 0$, show that $S_n \xrightarrow{D} s_0$ implies that $S_n \xrightarrow{P} s_0$.

Problem 2.5: A truncated Poisson distribution

Davison (2003) Section 2.2, Exercise 8.
Truncated Poisson variables Y arise when counting quantities such as the sizes of groups, each of which must contain at least one element. The density is

$$\mathrm{P}(Y = y) = \frac{\theta^y \mathrm{e}^{-\theta}}{y!(1 - e^{-\theta})}, \quad y = 1, 2, \ldots, \quad \theta > 0.$$

a. Find an expression for $\mathrm{E}(Y) = \mu(\theta)$ in terms of θ

b. If $Y_1, \ldots, Y_n$ is a random sample for this density and $n \to \infty$, give an argument for that $\overline{Y} \xrightarrow{P} \mu(\theta)$. Hence deduce that $\hat{\theta} = \mu^{-1}(\overline{Y}) \xrightarrow{P} \theta$.

Finally, an interesting application:

Problem 2.6: A model for rare infections spreading in families

[see Davison, 4.9: Problem 12.]{smallgrey}
The following problem looks longer than the original in Davison, only because it is partitioned into steps and slightly simplified, and contains hints:
A family has two children A and B. Child A catches an infectious disease D which is so rare that the probability that B catches it other than from A can be ignored.
Child A is infectious for a time U having probability density function

$$f_U(u) = \alpha \mathrm{e}^{-\alpha u},\ u \geq 0, \text{ where } \alpha > 0.$$

To model spreading the infection to child B one assumes that in any small interval of time $[t, t+\delta t]$ in $[0, U)$, child B will catch the disease from A with probability $\beta \cdot \delta t$, where $\beta > 0$.
Under these assumptions, the probability ρ that child B catches the infection from child A is given by

$$\rho = \frac{\beta}{\gamma}, \text{ where } \gamma = \alpha + \beta. \tag{1}$$

In a family, where B actually is infected, the time T from infection of child A to infection of child B has an exponential distribution with density

$$f_T(t) = \gamma \mathrm{e}^{-\gamma t}. \tag{2}$$

An epidemiologist observes n independent similar families, that is families with two children where at least one has been infected. In r of these families, the second child catches the disease from the first, at times $t_1, \ldots, t_r$

a) Write down the likelihood of the data as the product of the probability of observing r and the likelihood of the fixed sample $t_1, \ldots, t_r$.

b) Find the maximum likelihood estimators $\hat{\rho}$ and $\hat{\gamma}$ of ρ and γ, and the asymptotic variance of $\hat{\gamma}$.

c) Find the maximum likelihood estimators for the original parameters α and β from $\hat{\rho}$ and $\hat{\gamma}$.

Fan Corner Mathematical Modelling

For those of you, who are into mathematical modelling, you can solve the first to original sub problems from Davison: derive Equations (1) and (2).
There are different ways to get to these results. You could solve the differential equation that is implied in the model, or follow the hints below:

i) To calculate the probability that B does *not* catch the infection within the interval $[0, u)$, set $\delta = u/m$ for $m \in \mathbb{N}$. Then write the interval $[0, u)$ as union of m intervals of length δ. Let m go to infinity.

ii) One way to find the distribution function of time T to infection for child B in the family where A and B have got the disease would again be to first calculate the probability that B does *not* get infected in the interval $[0, t]$. You could separately consider cases where $U < t$ and where $U > t$, but perhaps you also have a smarter idea. For the final result, remember to divide by the probability that B gets infected, given A was infected.